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Research Report

Effects of tetramethylpyrazine on nitric oxide/cGMP signaling after cerebral vasospasm in rabbits

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ABSTRACT

Tetramethylpyrazine (TMP), an ingredient of Chinese herbal Szechwan lovage rhizome, shows vasorelaxant effect. Cerebral vasospasm (CVS) after subarachnoid hemorrhage (SAH) is associated with high mortality and morbidity. Here, we evaluated the effect of TMP in a model of CVS and sought to identify the underlying mechanisms of action. A rabbit SAH model was established by injection of the autoblood via cisterna magna. Cerebral blood flow and arterial diameter were measured by Transcranial Doppler (TCD) and Computed Tomography Angiography (CTA). Expression of eNOS and PDE-V in basilar artery (BA) was assessed by western blots. Levels of nitric oxide (NO) in plasma and cerebral spinal fluid, and of intra-endothelium Ca^{2+} were measured. Significantly reduced diameter and accelerated blood flow velocity were detected in BAs of SAH animals ($P<0.05$ vs. sham group). Expression of eNOS and NO was increased, and PDE-V expression was reduced by TMP.TMP ameliorated cerebral vasospasm ($P<0.05$ vs. SAH group), and L-NAME (a NOS inhibitor) partly abrogated the effects of TMP. TMP induced a dose-dependent increase of intra-endothelium Ca^{2+}. The current results demonstrated that the vasorelaxant effect of TMP was at least in part via regulation of NO/cGMP signaling.

1. Introduction

Cerebral vasospasm (CVS) following subarachnoid hemorrhage (SAH) is associated with high mortality and morbidity (Kolias et al., 2009; Pyne-Geithman et al., 2009a,b). It has been reported that multifactorial mechanisms were involved in the occurrence of CVS (Rothoerl and Ringel, 2007; Gokce et al., 2010; Güresir et al., 2010). Recent studies indicated that dysfunction of eNOS may contribute to the development of CVS (Pyne-Geithman et al., 2009a,b; Osuka et al., 2009). As the key enzyme in nitric oxide (NO)/cGMP signaling pathway, eNOS was activated by binding with intra-endothelial Ca^{2+} and calmodulin. Endothelial NOS synthesis NO, the latter diffuses rapidly into adjacent vascular smooth muscle cells and combines with histidine ligand at β1 subunit of SGC, leading to the transformation of GTP to cGMP and vasodilatation (Murad, 2006; Denninger and Marletta, 1999). There is a close relationship between NO production and attenuation of CVS, and studies have revealed that delivery of NO ameliorates post-SAH vasoconstriction (Pradilla et al., 2004). However,

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evidences show that NO replacements were not always valid in treatment of CVS as being expected (Raabe et al., 2002).It means that NO/cGMP signal may not be the sole factor that is responsible for CVS.

The current therapy of CVS mainly depended on "triple-H therapy" (hypertension, hypervolemia, and hemodilution) and/or dihydropyridine calcium antagonists (Otten et al., 2008). Nevertheless, new medicines and methods are presented in recent years. Tetramethylpyrazine (TMP, Fig. 1), the active ingredient of Chinese herb Szechwan lovage rhizome, has been reported to possess therapeutic effects on vertebrobasilar ischemia (Ge et al., 2008). Our previous research found that TMP ameliorated CVS in SAH rats through the regulation of caspase-dependent apoptosis (Gao et al., 2008). However, the effects of TMP on NO/cGMP signal after SAH were not elucidated. Here, the effects of TMP on CVS and NO/cGMP signal were explored in SAH models.

2. Results

2.1. TMP attenuates CVS after SAH

All animals survived during the experimental period. Respiratory arrest occurred in four rabbits (12.5%) during autoblood injection. No neurological deficits were observed after external cardiac massage.

As compared with sham-operated animals, significant increase of mean MFV and PSV was detected in basilar arteries of SAH animals ($P<0.05$; Fig. 2). TMP (30 mg/kg, i.v.) resulted in significant reduction in both MFV and PSV compared with that of SAH animals (Figs. 2E and F). Animals treated with both TMP and L-NAME (inhibitor of NOS) had mean MFVs and PSVs similar to that of SAH animals (Figs. 2E and F).

We measured BA diameters using CTA. The schematics of scanning locations are shown in Figs. 3A and B, and C. Two rabbits (6.25%) suffered unilateral pinnal hematoma resulted from involuntary movements. Neither neurological deficits nor deaths were observed during and after the performance of CTA. Representative images of CTA were shown in Figs. 3D–G. Bead-like images and significant reduction of BA diameter were detected in SAH group as compared with that of sham group ($P<0.05$; Fig. 3E and H).This confirmed successful establishment of CVS models. Compared with SAH group administration of TMP induced marked increase of BA diameter ($P<0.05$; Fig. 3H), while subsequent application of L-NAME attenuated the vasorelaxant effect of TMP ($P<0.05$ vs. TMP group; Fig. 3H).

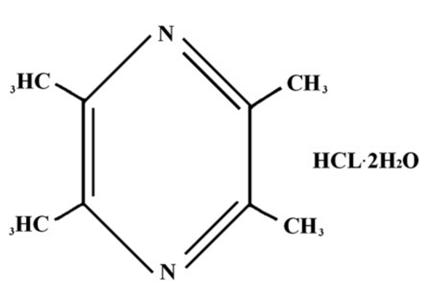

Fig. 1 – Chemical structure of tetramethylpyrazine hydrochloride (TMP).

2.2. TMP regulates NO/cGMP pathway

It was reported that TMP-induced relaxation in rat pulmonary artery was endothelium and NO-dependent (Peng et al., 1996), which prompted us to examine eNOS in basilar arteries of each group. As hydrolysis of cGMP by type-V phosphodiesterase (PDE-V) was the termination of NO/cGMP signal, PDE-V was determined as well. Western blot analysis of eNOS and PDE-V were shown in Fig. 4. In line with previous study (Shih et al., 2008), SAH caused significant reduction of eNOS when compared with the sham group in our study ($P<0.05$; Figs. 4A and C). TMP significantly increased the impaired expression of eNOS (Figs. 4A and C). In comparison to the results of TMP alone, L-NAME attenuated the effects of TMP by down regulation of eNOS ($P<0.05$; Figs. 4A and C). On the other hand, PDE-V expression was markedly increased in SAH group as compared with sham group. The level of PDE-V was reduced by TMP. In comparison to the results of TMP group, addition of L-NAME attenuated the effect of TMP by increasing PDE-V expression ($P<0.05$; Figs. 4B and D).

2.3. Identification of basilar arterial endothelium by immunocytochemistry and TEM

Freshly isolated endothelial cells exhibited a short, rounded appearance upon the attachment (Fig. 5A). After 5–7 days of incubation, the cells grew in confluent monolayer and became larger and polygonal. Factor-VIII related protein, an important antigenic marker of endothelial cells (Banerjee et al., 1985), was detected with immunological identification. About 89.13% of cultured cells were Factor-VIII positive as calculated from three independent immunohistochemical experiments (Figs. 5B–D). By using transmission election microscopy (TEM), the ECs were identified by the fusiform nucleus (lower arrow, Fig. 5E) and the flat body, as well as the sparing microvillus in cell membrane (upper arrow, Fig. 5E). Rod-shaped cytoplasmic inclusions (Weibel–Palade bodies), reported by Jaffe et al. (1973) to exist in endothelial cells, were observed in cytoplasm of cells evaluated (Fig. 5F).

2.4. TMP increased systemic NO production and endothelial NO production

As indicated by western blots, administration of TMP significantly increased the expression of eNOS. We therefore examined the levels of NO for further analysis of eNOS activity (Fig. 6). We found that the plasma NO was reduced in SAH group when compared with that of the sham ($P<0.05$). In contrast, NO levels increased in TMP group as compared with SAH group ($P<0.05$). Compared with TMP alone, L-NAME caused marked reduction of plasma NO ($P<0.05$). When compared with sham group, NO in CSF was reduced after SAH ($P<0.05$). The level of NO returned approximately to the value of sham group in animals treated with TMP ($P>0.05$). Animals treated with TMP+L-NAME had similar levels of NO in CSF compared with that of SAH group.

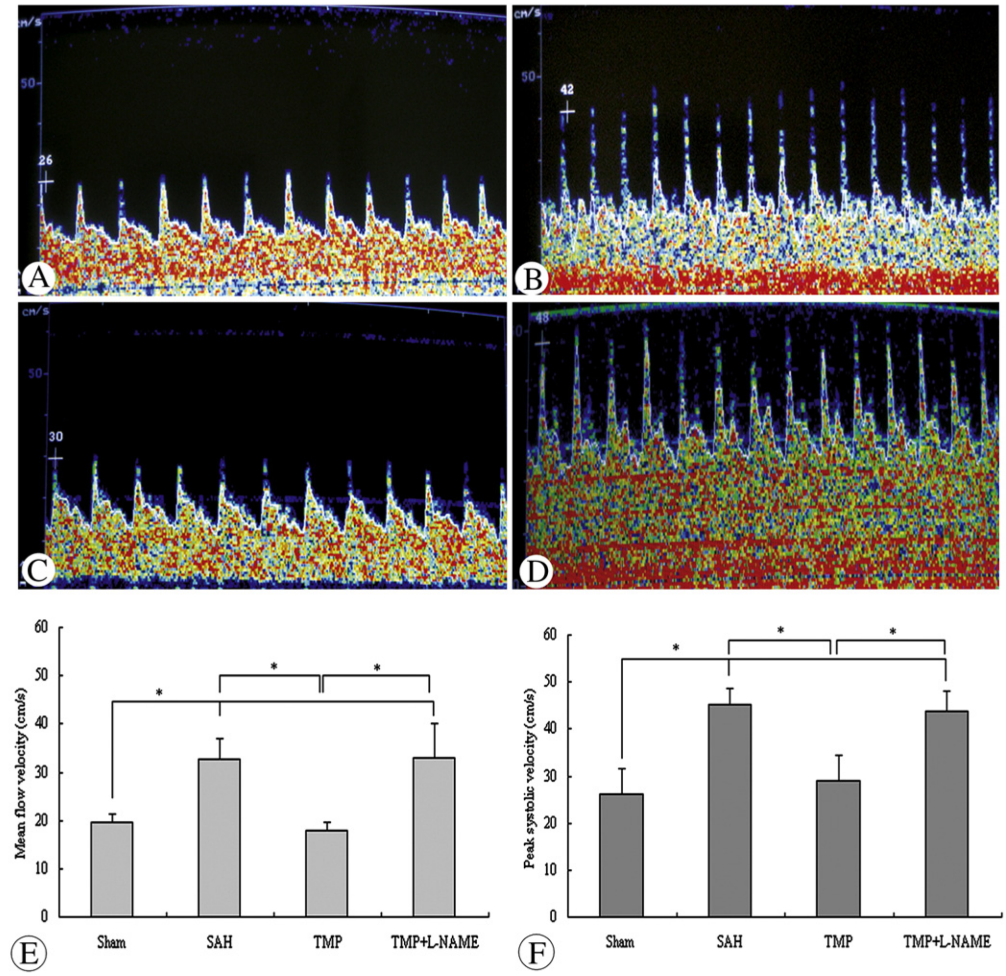

Fig. 2 – Measurement of cerebral blood flow. Representative TCD results of animals from A) sham, B) SAH, C) TMP, and D) TMP + L-NAME groups at day 3 after SAH. Compared with the sham group, mean MFV and PSV were significantly increased after SAH. In comparison to the SAH group, blood flow velocity was reduced after injection of TMP. Treatment of L-NAME abolished the effect of TMP, indicating a NOS-linked mechanism of TMP. E) Graphical representation of mean (±SD) MFV for all groups (*$P<0.05$, ANOVA; n = 8 each group). F) Graphical representation of mean (±SD) PSV for all groups (*$P<0.05$, ANOVA; n = 8 each group).

2.5. *TMP induces NO release and increases intracellular Ca^{2+} in vitro*

To prevent injury from overdose of TMP, cell viability was measured in preliminary experiments using Cell Counting Kit-8 (Dojindo laboratories, Kumamoto, Japan). Endothelial cells were treated with 0.01, 0.1, 1, and 10 mM TMP for 6, 12, 24 and 48 h, respectively. TMP treatment brought no impairments to cell viability at 6, 12 and 24 h post-TMP administration (data not shown). Endothelial cells were then treated with 0.1, 0.5, or 1 mM TMP. A dose-dependent release of NO was detected in cultural medium (Fig. 7). As Ca^{2+} is important for activation of eNOS, intra-endothelium Ca^{2+} was measured using laser confocal microscopy. The fluorescence increased with the increased application of TMP, which indicate that TMP dose-dependently enhanced Ca^{2+} uptakes in endothelial cells of BAs (Figs. 8A–E).

3. Discussion

In the present research, effects of TMP on CVS and its role in regulating NO/cGMP signal were detected after SAH. TCD was used for the assessment of real-time blood flow in BAs. According to Carrera et al. (2009) patients with elevated mean flow velocity more frequently suffered CVS on admission angiography compared to those with normal MFV. Vice versa, angiographic vasospasm predicted acute MFV elevation. In our study, the TCD results revealed marked increase of MFV and PSV in SAH group as compared with that of sham group. TMP significantly reduced both MFV and PSV. The TMP-induced reduction of MFV and PSV was abrogated by L-NAME, a NOS inhibitor, suggesting that the effect of TMP was NOS-dependent.

Our study was designed to investigate the underlying mechanisms of TMP on CVS resulted from SAH. Numerous

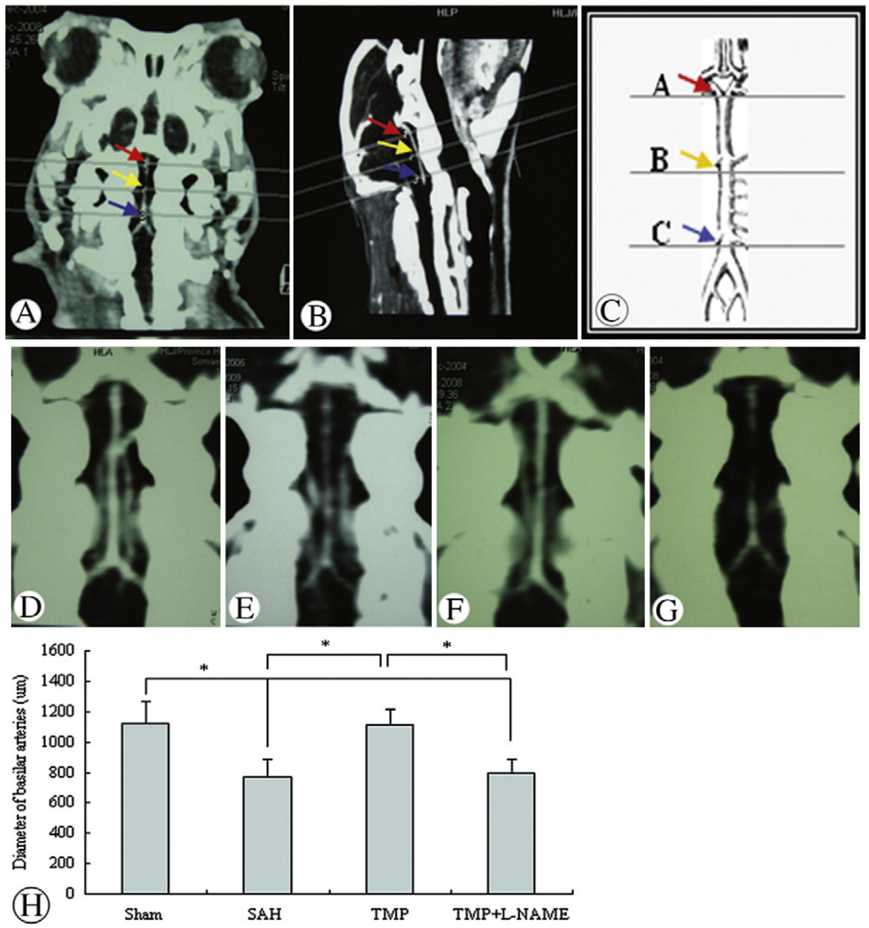

Fig. 3 – Measurement of BA diameter with CTA. A–C) Schematics of scanning locations at inferior of terminal bifurcation of the BA (upper arrow), superior of the vertebrobasilar junction (lower arrow), and midpoint of above two levels (middle arrow). Representative CTA images of animals from D) sham group, E) SAH group, F) TMP group and G) TMP + L-NAME group at day 3 after SAH. Vasospastic and bead-like images were detected in SAH animals. H) Graphical representation of mean (±SD) arterial diameters for each group (*$P < 0.05$, ANOVA; $n = 8$, each group). TMP significantly reduced post-SAH CVS, whereas the TMP-increased BA diameter was attenuated by concomitant application of L-NAME.

SAH models have been established to elucidate the mechanisms of CVS (Titova et al., 2009). Though the blood volume used varies with individual researchers, double-hemorrhage method is universally accepted for reliable induction of CVS in basilar arteries (Belen et al., 2007; Zhou et al., 2007; Yamaguchi Okada et al., 2009). In the present study, we carefully modified the double-hemorrhage method by boosting injected blood volume up to 1 ml/kg, so that intracranial hypertension mimicked a ruptured aneurysm in humans. During CTA performance, maximum intensity projection (MIP) technique was used to determine the degree of CVS. Significant reduction of mean arterial diameter was detected in SAH animals as compared with sham-operated controls. The results of CTA indicated successful establishment of animal models.. In accordance with our previous study on rats (Gao et al., 2008), TMP showed vasorelaxant effects on CVS animals. L-NAME, a NOS inhibitor (Niederhoffer and Szabo, 1999), partially attenuated the vasodilation induced by TMP.

Compared with sham group, SAH caused 72.6% and 35.6% reduction of NO in plasma and CSF, respectively. Treatment of TMP improved the low expression of NO to that of sham group, but the effect was abrogated when L-NAME was subsequently used. Endothelial NOS was proved a useful marker in predicting the development of CVS (Starke et al., 2008; Pyne-Geithman et al., 2009a,b). Our results confirmed the reports of Shih et al. (2008) that eNOS expression was reduced after SAH. TMP increased eNOS expression in SAH animals, and the effect was partially abolished by L-NAME. L-NAME is one of the most frequently used NOS inhibitors and effectively inhibits eNOS (Kalinowski et al., 2002; Sazonova et al., 2002; D'Souza et al., 2003; Peng et al., 2003).Our results, therefore, indicated that TMP-induced cerebral vasodilatation may be mediated by eNOS.

Several factors may affect post hemorrhagic levels of NO. Firstly, oxyhemoglobin released from blood clot envelopes the conductive arteries and scavenges NO. Secondly, destruction of nNOS-containing neurons can deprive the arteries of NO. Finally, increased level of endogenous inhibitor of eNOS, for example ADMA, as well as PDE-V inhibits eNOS function (Pluta and Oldfield, 2007). These factors lead to significant reduction of NO. Compared with SAH group, TMP significantly enhanced the levels of eNOS and NO. These observations indicated that

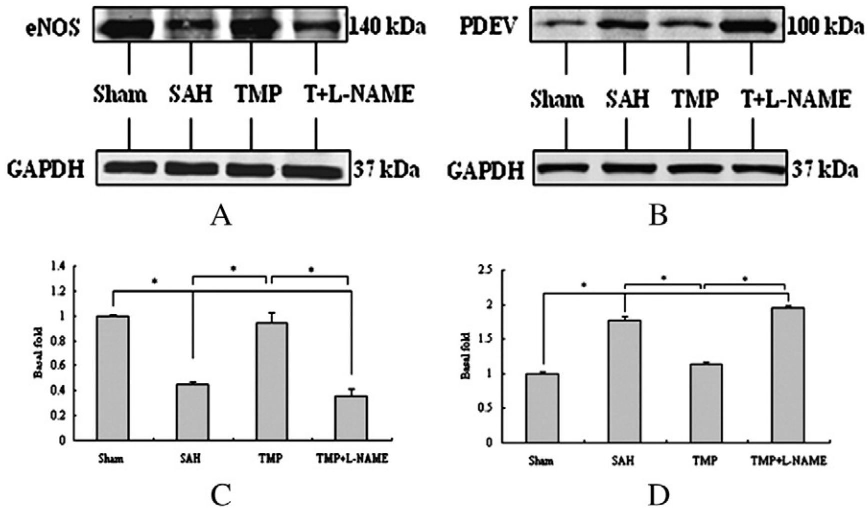

Fig. 4 – Western blot analysis of eNOS and PDE-V expression in basilar arteries. Western blots of A) eNOS and B) PDE-V in representative samples from animals from all groups. Graphical representation of mean ± SD of C) eNOS levels and D) PDE-V levels for each group (*$P < 0.05$, ANOVA; n = 4, each group). TMP reversed the impairments of SAH with eNOS and PDE-V expression in TMP group similar to that of sham-treated group. L-NAME attenuated effects of TMP by down regulation of eNOS and up regulation of PDE-V.

the up regulation of eNOS by TMP may be responsible for the elevated levels of NO. Nevertheless, influences from other isoforms of NOS cannot be absolutely ruled out based on present results. For example, Liu et al. (2010) reported that TMP inhibited LPS-induced over-expression of iNOS in N9 cells, and the NO level was decreased accordingly (Liu et al., 2010). While in our study the NO was increased after TMP treatment.

More than 10 isoforms of cyclic nucleotide phosphodiestrase (PDE) has been discovered. Type-V PDE is the isoform primarily responsible for the hydrolysis of cGMP (Inoha et al., 2002). As observed in canine models (Inoha et al., 2002) PDE-V was over-expressed in SAH rabbits as compared with sham-operated rabbits in present study. PDE-V expression was significantly decreased in TMP group when compared with that of SAH group. The TMP-induced expression of PDE-V seems contrary to the

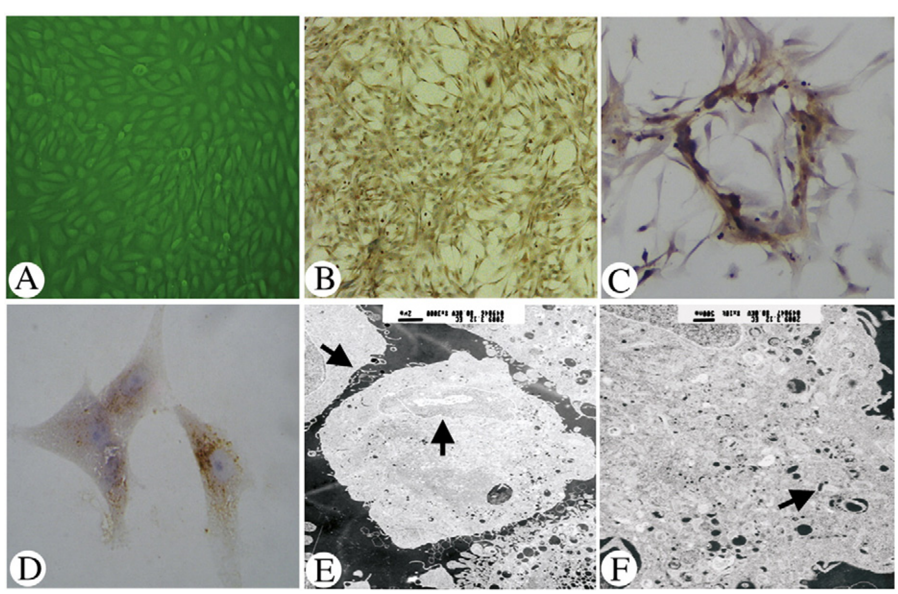

Fig. 5 – Characterization of endothelial cells isolated from basilar arteries of rabbits. A) Representative image of primary culture from basilar arterial endothelium (10 × 10). Results of immunocytochemistry staining for Factor-VIII associated protein: B) 10 × 10, C) 10 × 25, and D) 10 × 40. E) By transmission election microscopy, ECs were identified by the fusiform nucleus (lower arrow, × 3000) and flat body, as well as sparing microvillus in cell membrane (upper arrow, × 3000). F) A rod-shaped cytoplasmic inclusion (Weibel–Palade bodies), which typically exists in cytoplasm of endothelial cells, was indicated by the black arrow (× 10 k).

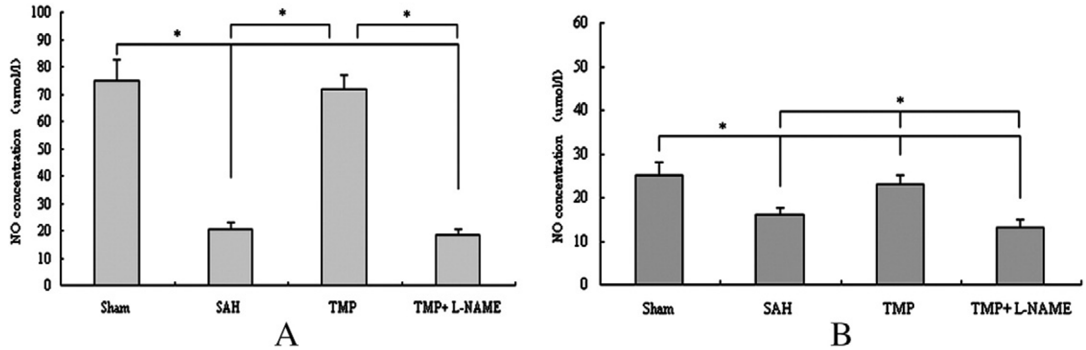

Fig. 6 – Measurement of NO levels in A) plasma and B) CSF (*$P<0.05$, ANOVA; n=8, each group). The NO levels were significantly reduced in SAH group as compared to sham group. TMP-treated animals had significantly higher levels of NO than that of animals in SAH group; effect of TMP was attenuated after the co-application of L-NAME.

established theory that eNOS facilitates the production of cGMP and the latter normally activates PDE-V. However, there is evidence that mRNA and protein expression of PDE-V were modestly increased in L-NAME-treated mesenteric arteries (Zhang et al., 2009). We observed similar but statistically significant increase of PDE-V in animals treated with TMP and L-NAME (Fig. 4D). To evaluate the effect of L-NAME alone, a second group of SAH animals (n=7) and SAH animals treated with L-NAME (n=8) were evaluated. We observed slight reduction of arterial diameter and increased MFV and PSV in L-NAME alone group ($P>0.05$, data not shown). In addition, the levels of NO in plasma and CSF were reduced after L-NAME treatment. Compared with SAH group, there was a moderate but significant reduction of eNOS expression in L-NAME group, and slight enhancement of PDE-V in basilar arteries (data not shown). Effects of L-NAME on CVS and expression of NO was then ruled out.

The Ca^{2+} inside endothelial cells sustains the activity of eNOS. Perivascular oxyhemoglobin presented after SAH may deprive ECs of intracellular Ca^{2+} through inactivation of Ca^{2+} channels and pumps (Scharbrodt et al., 2009), which lead to lowered expression of eNOS and dysfunction of NO/cGMP signal. After TMP treatment, the intra-endothelium Ca^{2+} was increased in a dose-dependent manner. To evaluate activity of eNOS, NO level was measured in culture medium. The

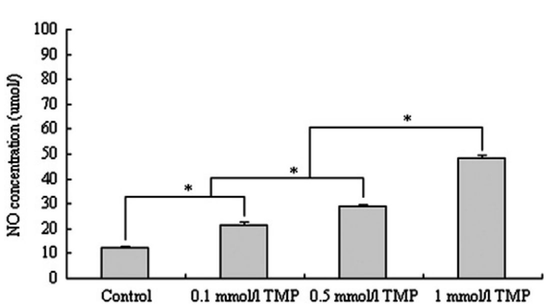

Fig. 7 – Measurement of NO released by cultured endothelial cells. To evaluate activity of eNOS we examined NO, the downstream product of eNOS, by treating cultured endothelial cells with 0 (control), 0.1, 0.5, and 1 mM TMP. Dose-dependent release of NO was detected in culture medium (*$P<0.05$, ANOVA; n=4, each group).

increase in NO by TMP further supported our hypothesis that the vasorelaxant effects of TMP were mediated partially through regulation of NO/cGMP signal. Intra-endothelial Ca^{2+} might act as the initial trigger. These observations indicate that regulation of intra-endothelium Ca^{2+} should be a useful approach in CVS therapy.

4. Experimental procedures

4.1. TMP

Tetramethylpyrazine hydrochloride (TMP) for *in vivo* use was purchased from Medisan Pharmaceutical Co., LTD (Harbin, Heilongjiang, China; purity≥95%). The reference substance of TMP used for in vitro assays was obtained from National Institute for the Control of Pharmaceutical and Biological Products (Peking, China; purity≥99.5%).

4.2. Animal model of cerebral vasospasm

Thirty-two male Japanese white rabbits, weighing between 2.3 kg and 2.6 kg, were used. All procedures followed the Guidelines for the Care and Use of Laboratory Animals published by the U.S. National Institutions of Health (NIH Publication No. 85-23, revised 1996) and were reviewed and approved by the ethics committee of animal experiments at Harbin Medical University. Animals were randomly divided into four groups. SAH was established by injection of autoblood via cisterna magna essentially as described previously (Pradilla et al., 2004; Marbacher et al., 2008). Briefly, rabbits were anesthetized with pentobarbital sodium (30 mg/kg, i.v.). A total of 1 ml of cerebrospinal fluid (CSF) was withdrawn using a 25-gauge needle and the atlanto-occipital membrane was exposed. Subsequently, 1 ml/kg unheparinized autoblood from an auricular artery was injected into cisterna magna at a rate of 1 ml/min. Then, the animal was restrained in a head-down position (30°) for 30 min to allow the blood diffuse to subarachnoid space. To induce delayed cerebral vasospasm, autoblood injection was repeated after 48 h. Throughout the operation, physical variables were monitored by exposing and placing a catheter through right femoral artery essentially as described by our previous report (Gao et al., 2008), and rectal

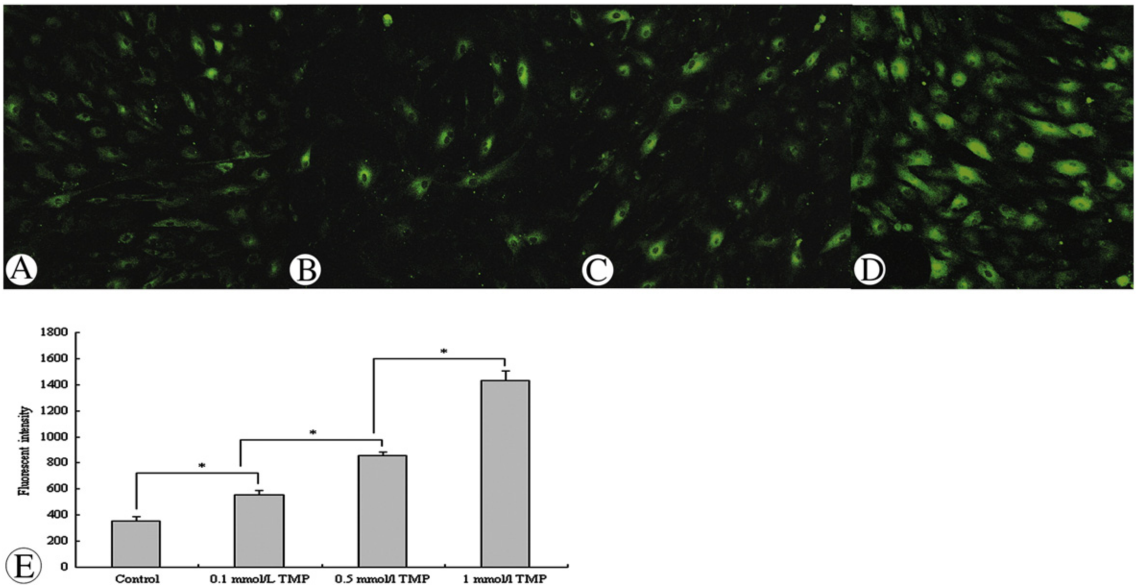

Fig. 8 – Detection of intracellular Ca^{2+} with laser confocal microscopy. Representative images from endothelial cells treated with A) 0, B) 0.1 mM TMP, C) 0.5 mM TMP, and D) 1 mM TMP. E) Mean fluorescent intensity in each group was measured 12 h after addition of TMP (*$P < 0.05$, ANOVA; n = 20 each group).

temperature was maintained at 37 °C using heating pads. Sham operation was performed with the same procedure except that saline was injected instead of autoblood. For TMP group, 30 mg/kg TMP in saline was administered intravenously 30 min after SAH induction and then daily for 3 days; the sham and SAH groups were injected with the same volume of saline. Animals in the TMP+L-NAME group received L-NAME, a NOS inhibitor, (30 mg/kg, i.v., Sigma Aldrich) in saline after TMP administration. Since most CVS in rabbits occurs 3 days after autoblood injection (Pradilla et al., 2004), animals were sacrificed on day 3 for imaging and biochemical analysis.

4.3. *Measurement of blood flow velocity and diameter in basilar artery*

To avoid hemodynamic disturbance by contrast injection, Transcranial Doppler sonography (TCD) was performed prior to Computed Tomography Angiography (CTA). Rabbits were under general anesthesia (ketamine, 50 mg/kg, and xylazine, 20 mg/kg, i.m.) during the procedure. Mean flow velocity (MFV) and peak systolic velocity (PSV) in basilar artery were recorded on day 3 using TC 2000 (EME, Ueberlingen, Germany). A 4-MHz probe with pulse-wave mode was used transoccipitally and angled until the clearest blood signal was detected (Göksel et al., 2001). The basilar artery was investigated using CTA (Laslo et al., 2008). Rabbits were restrained in a prone position with medical tape to minimize motion artifacts. A set of non-contrast axial images was obtained to verify head position and scanning locations. A total of 5 ml of omnipaque (300 mg/ml; Nycomed, Peking, China) was intravenously delivered using high-pressure syringe (0.5 ml/s), and the non-contrast scout image was acquired on Somatom Sensation 16 (SIMENS Company, Germany). The scanning parameters were as follows: 120 kV and 80 mA; 512×512 image frame; 2-mm slice thickness; 18 s delay time between the start of contrast administration and of scanning. Arterial diameters were averaged at three levels: inferior of terminal bifurcation of BA, superior of the vertebrobasilar junction, and midpoint of the above two levels. Independent investigator blinded to experimental protocols analyzed the experimental data.

4.4. *Western blot analysis*

On day 3, basilar arteries were removed and frozen in liquid nitrogen and stored at −80 °C. Basilar arteries from two rabbits from the same group were pooled for each assay. Vascular tissues were homogenized and protein concentrations were determined with BCA protein assay kit (Beyotime Institute of Biotechnology, JiangSu, China), following the manufacturer's instruction. A total of 50 μg protein was separated on a 10% SDS-polyacrylamide gel and transferred onto nitrocellulose membrane. The membranes were blocked by 5% blocking buffer (nonfat dry milk in 0.05% PBST) for 2 h at room temperature and incubated overnight at 4 °C with mouse monoclonal anti-eNOS (1:1000, BD Biosciences, San Jose, CA, USA) and rabbit anti-PDE-V (1:500, Cell Signaling Technology, Danvers, MA, USA) antibodies. After washing in PBST, the blots were incubated with biotin-labeled secondary antibodies (1:10,000, LI-COR Biosciences, Lincoln, NE, USA). Mouse anti-GAPDH monoclonal antibody (1:200, Santa Cruz Biotechnology, Santa Cruz, CA, USA) was used as internal control. Immunoreactivity was detected using Odyssey Imaging System (LI-COR Biosciences).

4.5. *Isolation, culture, and identification of basilar arterial endothelium*

Basilar arteries from untreated rabbits of 5–6 weeks of age were harvested after decapitation under sterile conditions. Endothelial cells (ECs) were isolated and cultured essentially as described by Ager (1987) and Fehrenbach et al. (2009) with some modifications. Briefly, the arteries were carefully cleaned of connective tissue, rinsed in ice-cold PBS (pH 7.35) containing penicillin (100 U/ml) and streptomycin (100 μg/ml), and cut into small segments in serum-free endothelial cell medium (ECM medium: 5% fetal bovine serum, endothelial cell growth supplement and penicillin/streptomycin; SciencCell Research Laboratories, Carlsbad, CA, USA). After centrifugation at 1000 rpm for 10 min, remaining tissue fragments were suspended and digested with 0.1% collagenase type I in 1% bovine serum albumin at 37 °C for 90 min with constant shaking. After enzymatic digestion, the tissue suspension was filtrated through 200-mesh screen and centrifuged at 1000 rpm for 10 min. Cells were then harvested and seeded into 25 cm^2 culture flasks, supplemented with complete ECM medium. ECs were identified using transmission electron microscopy as described by Larson and Sheridan (1982) and were stained for with antibody to Factor-VIII associated protein (1:60, Zhong Shan Golden Bridge Biotechnology, Peking, China) as described by Garrafa et al. (2006). After 2 or 3 passages, ECs were seeded onto 1-cm^2 sterile cover slips in 6-well plates. After incubation at 37 °C for 12 h, cells were treated with 0.1, 0.5, or 1 mM TMP in serum-free ECM for 12 h. Culture medium was collected for nitric oxide assay, and cells attached on cover slips were used for fluo 3/AM staining.

4.6. *Nitric oxide assay*

NO in samples from auricular arterial plasma and CSF, collected on day 3 before the sacrifice of rabbits, and from culture medium of ECs treated by 0.1, 0.5 and 1 mM TMP were measured using NO assay kit following the manufacturer's instructions (Nanjing Jiancheng Bioengineering Institute, JiangSu, China). Serum-free ECM was used as base line. Absorbance at 550 nm was detected using UV-265 spectrophotometer (Shimadzu Corporation, Kyoto, Japan).

4.7. *Measurement of intracellular Ca^{2+} with laser confocal microscopy*

Briefly, cells attached on cover slips were treated by 0.1, 0.5 or 1 mM TMP for 12 h. Afterwards, cells were washed with Ca^{2+}-contained Tyrode's solution (135 mM NaCl, 5.4 mM KCl, 1.0 mM $MgCl_2$, 0.33 mM NaH_2PO_4, 5 mM glucose, 10 mM HEPES, 1.0 mM $CaCl_2$, pH 7.4) and incubated with 5 μM fluo 3/AM (Molecular probes, Eugene, OR, USA) containing 0.1% pluronic F-127 (Molecular probes, Eugene, OR, USA) at 37 °C in 5% CO_2 for 35 min in the dark. The mean fluorescent intensity was detected from twenty ECs in each group using laser confocal microscopy (Olympus, Tokyo, Japan). The fluorescence intensity was quantified with FV-1000 viewer.

4.8. *Statistical analysis*

One-way ANOVA was used for data analysis, and multiple comparisons were corrected with Student–Newman–Keuls test. Results with $P<0.05$ were considered significant. Data are presented as means ± standard deviations (SD).

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